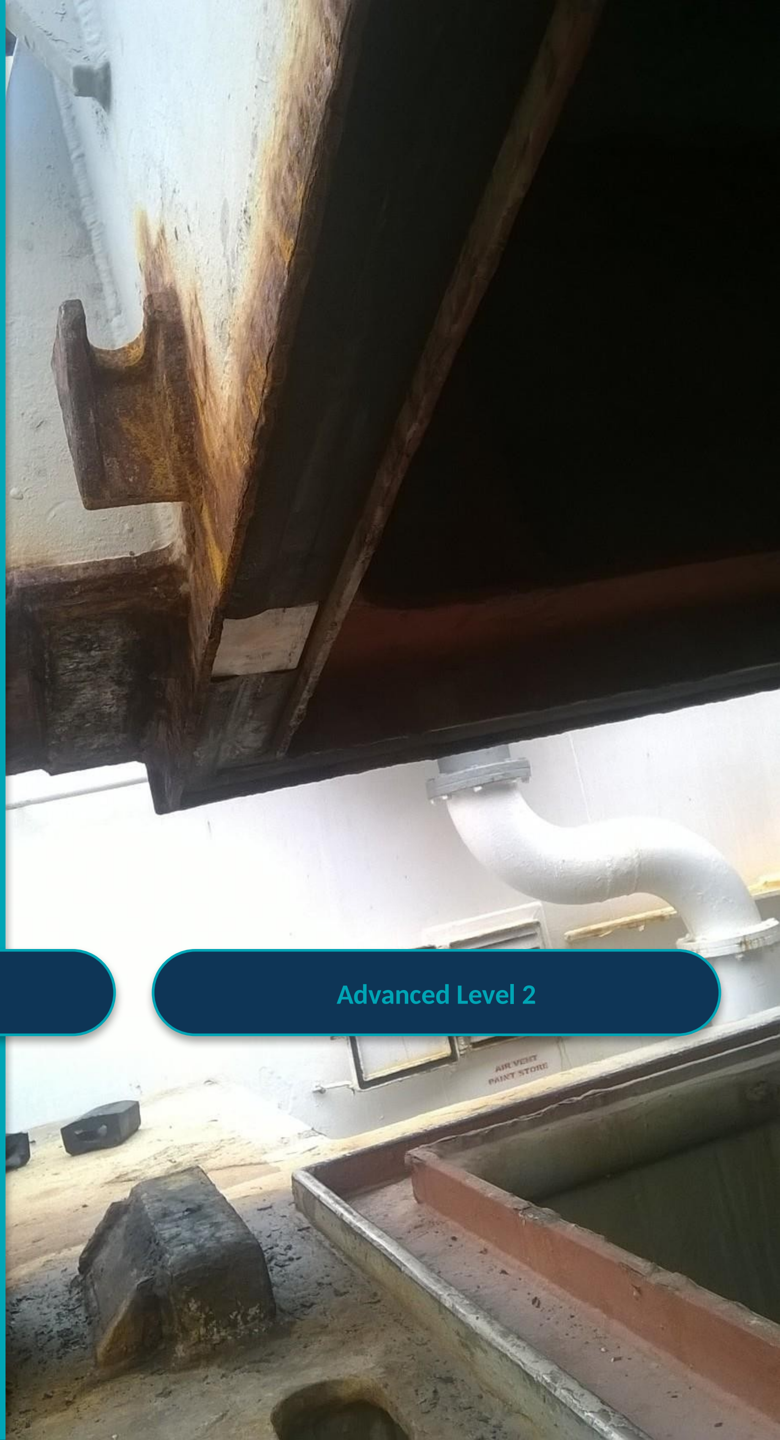
TECHNICAL & DRYDOCK MASTERCLASS

Hatch Cover Operations & Maintenance

The complete masterclass: structure, pads, sealing, mechanical and hydraulic systems, routine maintenance, and full drydock repair practice for hatch covers on bulk carriers



Class Rules

Load Line Convention

Company SMS / PMS

Advanced Level 2

100+ slides · 26 field photographs · Real worked examples & drydock figures

COURSE MAP

What we will cover — 14 sections

SECTION 01

Hatch Covers: Important, Complex, Difficult

Three words the original masterclass uses to frame this entire course.

Important

Hatch cover integrity affects cargo, crew and hull safety on every single voyage.

Full of Equipment

Steel structure, pads, seals, mechanical fittings and a complete hydraulic system, all in one assembly.

Genuinely Difficult

Precision fits, heavy components and hidden wear make this harder than it looks from the deck.

?

Everyone knows hatch covers matter — but too often nobody takes responsibility until something leaks. That has to change on this vessel.

SECTION 01

Precision Matters — The Cooking-Pot Analogy

A simple kitchen comparison the masterclass uses to explain why panel-to-coaming fit must be exact.

Lid Deformed

A warped lid never seals no matter how hard you press it — just like a warped panel.

Pot Deformed

A distorted rim leaks even under a perfect lid — just like a distorted coaming.

Lid Far Away

A lid that never reaches the rim seals nothing — like a panel resting incorrectly.

Correct Placement

Only a true lid on a true rim, correctly placed, seals every time.

SECTION 01

Why Hatch Cover Integrity Matters

A hatch cover that is not properly maintained can fail in more ways than one — and the consequences reach well beyond the cargo hold.

Side-rolling hatch covers on deck. Every panel, pad, wheel and hinge shown here is only as reliable as the last maintenance round performed on it.

Water Ingress & Cargo Damage

A cover that is not weathertight lets seawater into the hold — wetting cargo and triggering claims and corrosion.

Structural & Steel Damage

Worn pads, misaligned wheels and deteriorated rubber accelerate wastage of coaming and panel steelwork.

Off-Hire & Detention

A failed weathertightness test or defective securing device can trigger PSC detention or costly off-hire.

Crew Safety

Hydraulic pressure, heavy panels and seized hinge pins make this some of the highest-risk work on deck.

Types of Hatch Covers

The three main designs found on general cargo and bulk carriers, sized by length, width, height and weight.

Type	Length (m)	Width (m)	Height (m)	Weight (t)
Side Rolling	14 – 18 (typ. 16)	8 – 18 (typ. 10)	0.8 – 1.5 (typ. 1.0)	20 – 40 (typ. 25×2)
Folding / Multifolding	12 – 20 (typ. 16)	6 – 12 (typ. 10)	0.9 – 1.3 (typ. 1.0)	20 – 40 (typ. 30×4)
Lift-Away	10 – 28 (typ. 16)	5 – 13 (typ. 13)	0.7 – 1.2 (typ. 1.0)	10 – 50 (typ. 25)

Matching Design to Vessel Size

The three designs are not interchangeable — each is matched to a hold-opening size and operating profile.

- Minibulkers, Handysize and Handymax vessels are typically fitted with folding or multifolding hatch covers, well suited to smaller hold openings.
- Panamax and Capesize vessels predominantly use side-rolling covers, giving the wide clear openings these hulls require.
- VLBCs (Very Large Bulk Carriers) often use a single end- or side-rolling panel per hold, sized for the largest hatch openings afloat.
- The bigger the hold opening, the more the design has to resolve stacking, sealing and structural load together — there is no free choice at large sizes.

Side-rolling covers on a Panamax-class deck.

SECTION 02

Stacking vs. Weathertightness — The Design Trade-off

Every hatch cover design balances two competing goals at once.

Easier Stacking

Designs optimized for fast, simple opening and stowage of panels during cargo operations.

Easier Sealing

Designs optimized for a simpler, more reliable weathertight closure once the panel is shut.

No single design wins on both counts — the right choice balances stacking speed against sealing reliability for the vessel's trade.

SECTION 02

Types of Pot — Panel Seating by Design

The 'pot' is the recessed seat where a panel's resting pad lands on the coaming — its shape follows the hatch cover type.

VLBC — A single wide panel typically seats across one long pot run the full width of the hatch.

Side Rolling — Each panel seats on its own pot section along the coaming as it rolls into the closed position.

Folding — One Panel — A folding cover's end panel seats on a pot section shaped for its fold and hinge line.

Always confirm the actual pot geometry against the vessel's own drawings — never assume one hatch cover's layout matches another.

SECTION 03

Steel Structure: Open-Web vs Box-Type

Every hatch cover panel is built as either an open-web or a box-type steel structure.

Open-Web Construction

Lighter framing with open web girders — common on smaller panels, easy to inspect internally.

Box-Type Construction

Fully enclosed box girders for greater stiffness on larger, heavier panels.

SECTION 03

Open Construction vs Closed Construction

Beyond girder type, every panel is also built as either an open or a fully closed structure.

Open Construction

Structural members are open-profile, letting any water that gets past the seal drain away rather than collect inside the panel.

Closed Construction

Fully boxed and typically sealed, giving greater stiffness but requiring careful internal corrosion protection.

SECTION 03

The Four Critical Component Groups

Every hatch cover, whatever its type, is built from the same four functional groups.

Pads & Stoppers — Carry panel and cargo weight, transferring load into the coaming at each resting position.

Sealing & Drain — Rubber compression plus drain channels and valves — the weathertight barrier itself.

Mechanical — Cleats, wheels, hinges and chain — securing the panel closed and moving it open and shut.

Hydraulics — Power unit, pipes, controls, cylinders and motors that drive the covers.

SECTION 03

Workshop 1A — Find the Clearance Between Hatch Cover and Coaming

A practical drawing-and-verification exercise before touching any tool.

WORKSHOP EXERCISE

- Using the vessel's hatch cover general arrangement drawing and the manufacturer's manual, locate the designed clearance between the panel and the coaming.
- Read the drawing carefully and note the specified value.
- Then check that value against the actual condition on deck — drawings and reality do not always agree, and the difference is exactly what you are looking for.

How Pads Carry Load

Every resting pad and every metre of perimeter rubber is sized to one simple balance of forces.

$$\Sigma \text{ Weight (cargo + hatch cover)} = \Sigma \text{ Vertical Reactions (pads)} + (\text{Rubber Compression Reaction} \times \text{Length})$$

- The combined weight bearing down on the panel must be matched, point for point, by the upward reaction of the resting pads and the compression reaction of the perimeter rubber.
- If a pad is worn low or missing, the rubber is left to carry load it was never designed for — and it fails.
- This is exactly why pad condition is checked before rubber condition at every inspection.

Pads Evolution: Two Generations of Design

How resting-pad technology changed over four decades.

1970s–1990s

First Generation

Fixed steel-to-steel resting pads with limited wear allowance, requiring labour-intensive, infrequent replacement.

2000s–2010s

Second Generation

Modern replaceable pad systems introduced bronze, plastic (Polypad) and rubber (Flexipad) wear surfaces for longer life and simple renewal.

SECTION 04

Contact-Surface Pairings in Use Today

Four pairings account for almost every resting pad found in service.

Steel to Steel

The original pairing — highest wear rate, needs the tightest inspection interval.

Steel to Bronze

Improved wear resistance and a smoother running surface than steel-on-steel.

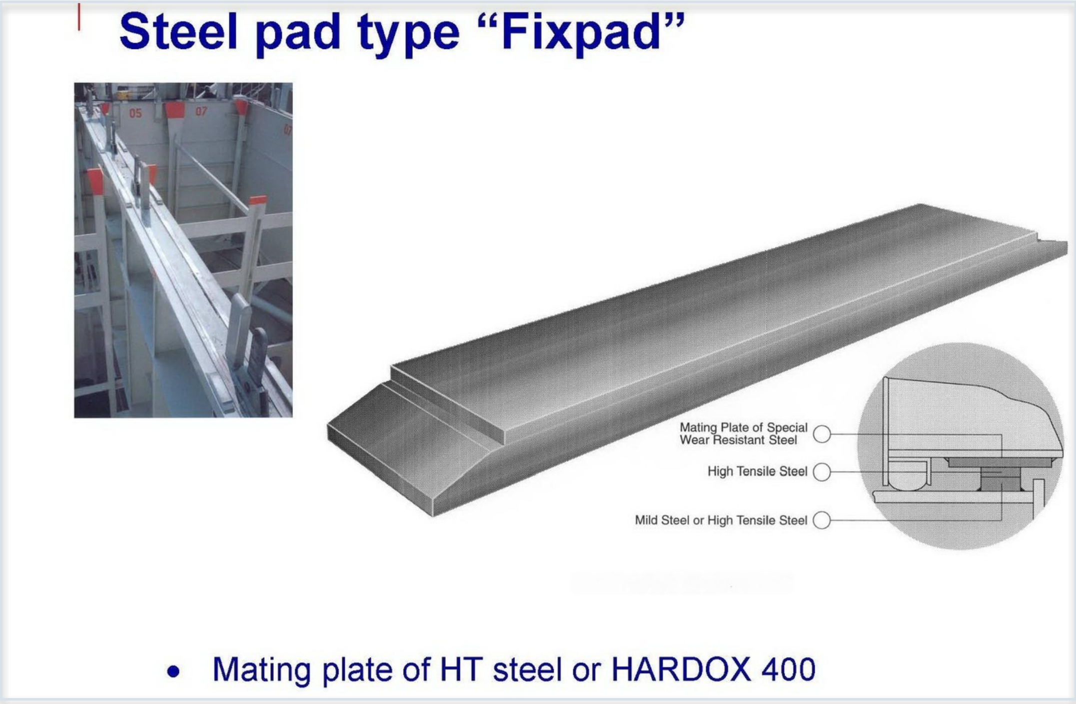
Steel to Plastic (Polypad)

A composite wear surface designed for long life and straightforward renewal.

Steel to Rubber (Flexipad)

Adds a degree of resilience and cushioning to the resting-pad contact.

Fixpad — Fixed Steel Pad



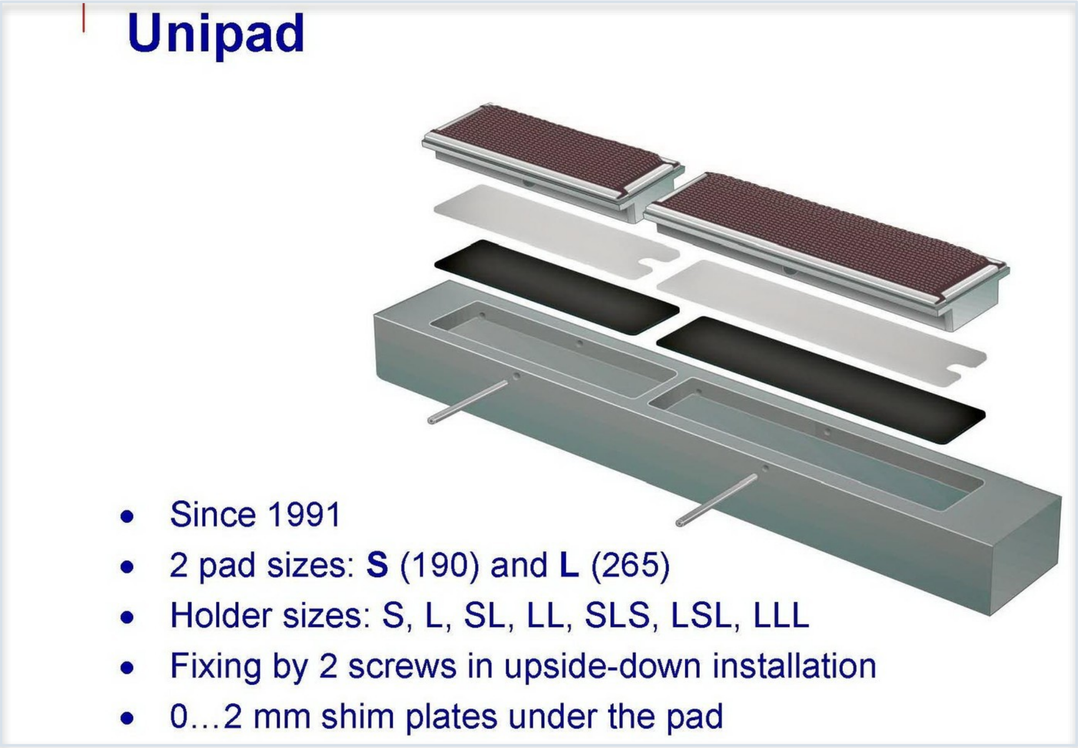
- A fixed steel resting pad — the traditional steel-to-steel design still found on many older panels.
- The wear surface is not renewed on its own; the mating plate must be monitored and replaced as one piece once worn.
- Refer to the maker's drawing (shown) for the exact plate material and mounting dimensions.
- Because renewal means removing the full plate, inspect Fixpad positions early in the preparation stage so replacement steel can be ordered in time.

SECTION 04

Replaceable Steelpad

- A steel pad designed to be unbolted and renewed without cutting or welding into the panel structure.
- Significantly reduces drydock time compared with a welded fixed pad.
- Still a steel-to-steel contact, so it carries the same wear rate as Fixpad — the advantage lies purely in how quickly it can be changed.
- Confirm bolt torque and mounting condition, not only plate wear, at every inspection.

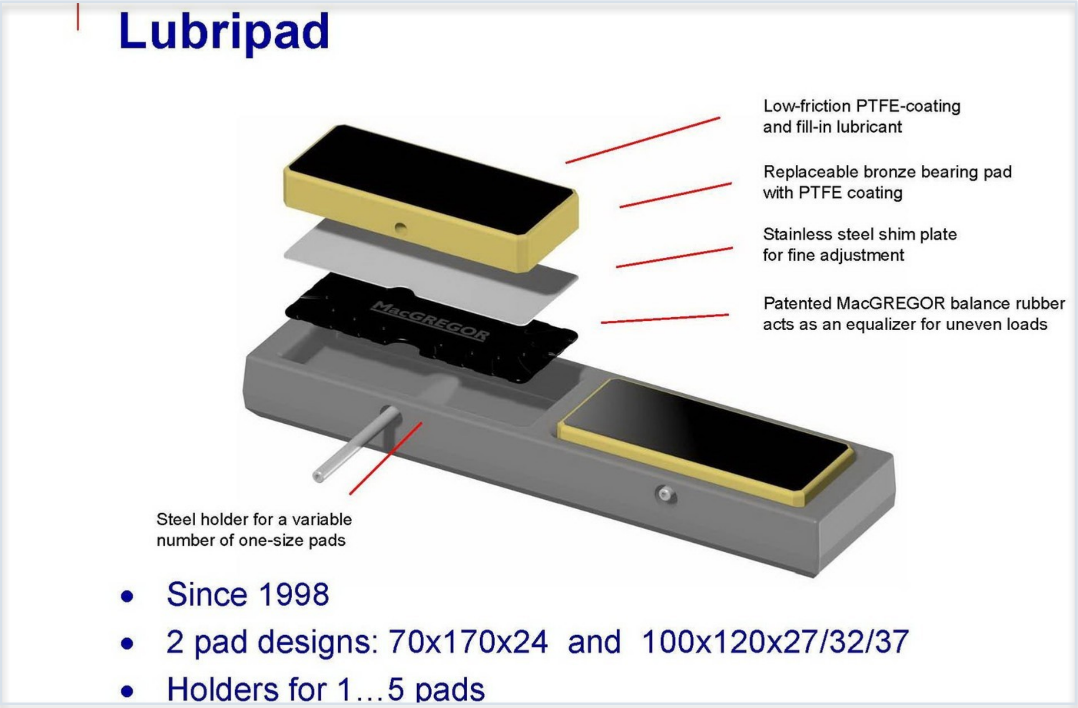
Unipad — Universal Design



- A universal-design pad intended to fit a standardised range of panel and coaming geometries.
- Simplifies spares holding, since one design can cover multiple resting-pad positions on the vessel.
- Check that the universal fit is genuinely correct for each position — 'nearly fits' is not the same as fits.
- A good candidate for standardising spares across a fleet of sister vessels.

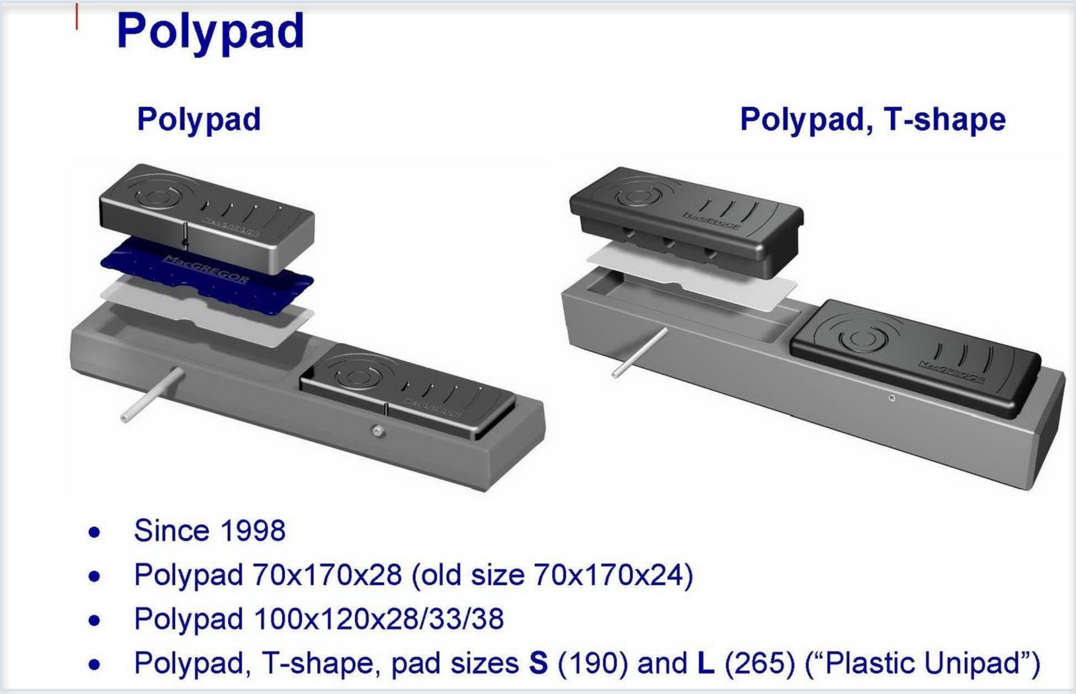
Lubripad — Self-Lubricating

- A self-lubricating pad surface that reduces friction and wear during panel operation.
- Particularly suited to positions that see repeated sliding contact rather than a single static resting load.
- Requires no external greasing at the pad face itself — but surrounding mounting hardware still needs the routine lubrication schedule.
- Monitor for glazing or surface hardening over time, which reduces the self-lubricating effect.



SECTION 04

Polypad — Steel-to-Plastic



- A steel-to-plastic composite pad, part of the modern replaceable-pad generation.
- The plastic wear surface is renewed on its own, without disturbing the steel backing plate.
- Good wear resistance for the renewal effort involved, which is why it appears widely in current pad specifications.
- Check for UV and chemical degradation of the plastic surface, not only mechanical wear.

SECTION 04

Flexipad — Steel-to-Rubber

- A steel-to-rubber pad that adds a degree of resilience and cushioning to the resting-pad contact.
- Helps absorb shock loading compared with a rigid steel-to-steel or steel-to-bronze contact.
- Inspect the rubber element for compression set and cracking, in addition to checking wear.
- A useful choice at positions that see uneven or impact loading rather than a smooth, steady contact.

SECTION 04

Reading Pad Wear — What to Look For

A pad's wear pattern tells you far more than its remaining thickness alone.

A paw-shaped compression wear pattern on a resting pad.

An uneven or 'paw-shaped' wear mark like this usually points to panel misalignment or an overloaded pad position — investigate the cause, not just the symptom.

SECTION 04

Choosing the Right Pad Type — Summary

A quick reference across all four contact pairings.

Pairing	Wear Resistance	Renewal Effort
Steel to Steel	Lowest	High — full pad replacement
Steel to Bronze	Improved	High — full pad replacement
Steel to Plastic (Polypad)	Good	Low — wear surface only
Steel to Rubber (Flexipad)	Good, with cushioning	Low — wear surface only

Match the pairing to the position's load and access — not every resting pad on the panel needs the same solution.

SECTION 05

Rubber Compression Profiles in Use

Three standard rubber cross-sections referenced in the masterclass, identified by their compressed dimensions.

Profile (mm)	Typical Use
71 × 40	Standard perimeter sealing profile
94 × 50	Heavier-duty perimeter profile
72 × 67	Higher-compression profile

Always confirm the exact profile against the vessel's own rubber specification before ordering — profiles are not interchangeable between panels.

SECTION 05

Perimeter Sealing — One Continuous Barrier

The rubber run around a panel is only as strong as its weakest few centimetres.

- The perimeter rubber must form one continuous, unbroken sealing line around the full panel edge.
- Every joint, corner and end piece is a potential leak path if it is not fitted and glued correctly.
- A sealing arrangement is judged by its weakest few centimetres — not by its average condition.
- This is why corner and end-piece work gets more attention, not less, than the long straight runs.

SECTION 05

Pad & Stopper Arrangements Differ by Hatch Type

Side-rolling, sliding-wedge and folding covers each use a different pad and stopper geometry.

Side Rolling

Resting pads and stoppers positioned along the rolling track and at the closed position.

Sliding Wedge

A wedge-type stopper draws the panel down onto its seating as it reaches the closed position.

Folding

Pads and stoppers positioned at the fold line and at each panel's outer resting edge.

Always check the arrangement against the vessel's own drawings and actual condition — never assume one hatch cover's layout matches another.

Workshop 1B — Rubber Compression Pressure

A calculation exercise on the 92×50 profile.

WORKSHOP EXERCISE

- Calculate the pressure required to compress a 15-metre length of 92×50 rubber at a cross joint.
- Use the manufacturer's compression-force-per-metre figure for the 92×50 profile as your starting point.
- Compare your answer with the compression bar's actual clamping capacity at that joint.

SECTION 05

Worked Example — Rubber Reaction Force

The same load balance from Section 04, applied to one real folding hatch.

18m × 20m folding hatch, compression rate 3.8 kN per metre of rubber

- Perimeter: 18×2 + 20×2 = 76 metres of rubber around the panel edge.
- Perimeter force: 76 m × 3.8 kN/m ≈ 289 kN — about 29 tonnes force.
- Cross-joint (18 m): 18 m × 3.8 kN/m ≈ 68 kN — about 7 tonnes force.
- Every metre of rubber on the panel is doing real, calculable work — none of it is decorative.

SECTION 05

Drain Channels on the Coaming

The first line of drainage, built into the coaming itself.

- Channels run along the coaming to collect any water that gets past the perimeter rubber.
- Must be kept clear of cargo residue, rust scale and old sealant at every inspection.
- Feed drain valves that must open freely — a blocked drain turns a minor seal imperfection into standing water on the seal.
- Included in the routine maintenance schedule's cleaning and derust-and-paint steps, not treated as a one-off drydock item.

A five-minute check of the coaming drain channels during a normal round can prevent a hold-wetting claim months later.

Drain Channels on the Cover

A second, parallel drainage path built into the panel itself.

- Channels on the underside of the cover collect water on the panel side of the seal.
- Work together with the coaming-side channels — one system alone is not the complete design.
- Corrosion and rubber-channel edge damage here are exactly what the routine derust-and-paint step exists to catch.
- Inspect from underneath with the cover open — damage here is easy to miss from a normal deck-level walk-round.

Coaming channels and cover channels are one system with two halves — checking only one half is not a complete drainage inspection.

Achieving Weathertightness — The Two Systems

Weathertightness is not watertightness, and it does not depend on any mechanical or hydraulic component.

Rubber System

A compression bar acting on the perimeter rubber creates the primary weathertight seal.

Compression bar + rubber, working together.

Drain System

Drain valves and drain channels remove any water that does get past the rubber before it reaches the hold.

Drain valves + drain channels, working together.

Both systems must be checked together — a perfect rubber seal with a blocked drain still leaves standing water on the seal, and a clear drain cannot compensate for a poor rubber seal.

SECTION 05

Weathertightness Is Not Watertightness

Three rhetorical questions from the masterclass, answered plainly.

- Weathertightness depends entirely on pads, sealing and drains — no mechanical or hydraulic component is involved in making the seal.
- The weight of the panel alone is never enough to guarantee a weathertight closure.
- Cleats provide mechanical security once the seal is made — they are not what creates the seal.
- If a cover leaks, look first at pads, rubber and drains — not at the cleats.
- This is the single most misunderstood point in hatch cover maintenance, and the most common reason a leak repair fails the first time.

This one page settles more disputes on deck than any other in this course — keep coming back to it whenever a repair isn't holding.

SECTION 06

Mechanical Components — Overview

Four component groups secure the panel closed and move it open and shut.

- **Cleats — Manual or Auto** — Hold the panel firmly closed once the weathertight seal is compressed.
- **Wheels** — Support panel weight and allow it to roll into and out of position.
- **Hinges** — Carry folding-panel loads through a spherical bearing at the fold line.
- **Chain — Wire or Rack-and-Pinion** — Drives the opening and closing motion of the panel.

Quick-Acting Cleats

- Available in manual and automatic (power-assisted) versions.
- Hold the compressed seal in place — inspect the cam, pin and rubber pad on the cleat itself at every round.
- A cleat that will not close fully usually means the seal beneath it is over-compressed or fouled, not that the cleat itself is faulty.

SECTION 06

Wheels & Wheel Tracks

- Carry the full rolling weight of the panel as it moves along the track.
- Check freeplay, flange condition and wheel-track wear on every inspection round.
- A worn wheel accelerates wear on the track it runs on — replace before the damage spreads.

Location of Wheels

Where wheels sit on the panel affects how load and wear are distributed.

- Wheel positions are fixed by the original design — never relocate a wheel without an engineering review.
- End and intermediate wheel positions typically see different load patterns and should be inspected as separate items.
- Record wheel positions and condition individually in the equipment register, not as one combined line.
- A position that consistently shows heavier wear than its neighbours is telling you something about load distribution, not just about that one wheel.

Treat wheel location like any other maintained position — one that is 'always fine' usually just hasn't been checked recently.

Alignment in Wheels

Misalignment often shows up only at one end of the panel's travel.

- Misaligned wheels run unevenly on the track, accelerating both wheel and track wear.
- Check alignment with the panel closed and again with it fully open.
- Correct alignment before, not after, renewing a worn wheel — fitting a new wheel to a misaligned mounting only repeats the failure.
- A wheel that wears unevenly across its own tread, rather than all over, is almost always an alignment problem, not a material problem.

A five-minute alignment check between refits costs far less than an emergency wheel-track repair at sea.

SECTION 06

Hinges

- Carry folding-panel loads through a spherical bearing that must be greased at every point.
- Intermediate hinges are normally serviced with the hatch covers on shore; end hinges are usually serviced on site with the cover closed.
- Cylinder hinges carry an additional load path and deserve the same attention as the structural hinges.
- Discussion: if a hatch cover will not open fully, hinge condition is one of the first places to look.

SECTION 06

Alignment in Hinges

Clearance and noise, checked through the full cycle — not just at rest.

- Check for clearance through the full range of the panel's operating cycle, not just at rest.
- Listen for noise during operation — a hinge that is starting to bind announces itself before it seizes.
- Grease every hinge point on the schedule, not only the ones that are easy to reach.
- A hinge that is hard to reach is not a hinge that can be skipped — plan the round so every point actually gets greased.

Hinges rarely fail without warning — noise and clearance changes are the early signals this checklist is designed to catch.

Chain Systems — Wire & Rack-and-Pinion

- Wire chain and rack-and-pinion chain both drive the panel's opening and closing motion.
- Replace chain once wear exceeds 20% of thickness, or once correct tension can no longer be maintained.
- Check all chain wheels for wear alongside the chain itself — a worn wheel will damage a brand-new chain.

Mechanical Component Inspection Checklist

One combined round covers all four groups.

- **Cleats** — Cam, pin and rubber condition; confirm the full closing action.
- **Wheels** — Freeplay, flange wear, wheel-track condition.
- **Hinges** — Clearance through the full cycle, grease points, noise.
- **Chain** — Wear percentage, tension, chain wheel condition.

SECTION 07

The Hydraulic System — Five Subsystems

Five subsystems work together to open, close and hold the covers.

Power Unit
Drives system pressure.

Pipes
Carry oil to each panel.

Controls
Valves and levers.

Cylinders
Move each panel.

Motors
Drive chain and wheels.

None of the five works in isolation — a fault in one subsystem usually shows up first as a symptom in another. The next seven slides take each in turn.

Power Unit & Pump Checks

- The power unit drives system pressure for every cylinder and motor on the vessel.
- Routine checks: pressure, amperage, oil level, oil temperature, and condition of the last oil-tank clean plus filter change.
- A pump unit drawing higher amperage than normal for the same pressure is already telling you something is wearing.

Pipes

The distribution network carrying oil to every panel.

- Pipes carry pressurized oil from the power unit to the controls, cylinders and motors at each hatch.
- Long runs and exposed sections are the most common source of chafe damage and slow leaks.
- The newer type of pipe support (see Component Repairs) reduces chafe compared with older supports.
- Cleanliness matters as much as the pipe itself — contamination introduced at one joint travels through the whole system.

Pipe runs are cheap to inspect and expensive to replace in an emergency at sea — walk the full run, not just the visible sections.

SECTION 07

Controls

- Valves and levers direct oil to open, close and hold each panel.
- Leaks from control levers are a common early warning sign — trace them before they become a bigger repair.
- Reconditioning a valve is often more cost-effective than wholesale replacement, provided the body is not scored.

SECTION 07

Cylinders & Gasket Set

- Cylinders convert hydraulic pressure into the mechanical force that moves each panel.
- Every reseal covers four items together: the O-ring, oil seal, backup ring and dust / scraper seal.
- Protective wrapping, as shown, is standard practice for a cylinder awaiting reinstallation or overhaul.

SECTION 07

Hydraulic Motors

Drive chain and wheel movement on covers that use motor-driven opening.

- Check for leaks at the motor casing and shaft seal on every round.
- Replace the shaft seal proactively once any weeping is visible — waiting for a full leak means oil loss and possible contamination.
- A motor that runs hot or noisy under normal load should be inspected before the next voyage, not at the next drydock.

SECTION 07

Hydraulic Oil Specification

The oil itself is a critical component in its own right.

Parameter	Requirement
Temperature	Below 65 °C
Cleanliness	10 µm filtration
Viscosity	ISO Grade 32
Return-line filter	50 µm mesh

SECTION 07

Mechanical vs Hydraulic Components

Not every part of the system needs hydraulic power to do its job.

Mechanical Components

Cleats, wheels, hinges and manually-driven chain — can often be operated, inspected and repaired independently of the hydraulic system.

Hydraulic Components

Power unit, pipes, controls, cylinders and motors — depend on pressurized oil to function at all.

Workshop 1C — Locate Two Key Valves

A hands-on diagram exercise.

WORKSHOP EXERCISE

- Using the vessel's hydraulic diagram, locate the system pressure relief valve.
- Locate the warm-up valve and confirm what condition it should be in before operating the covers in cold weather.
- Confirm both valves against the physical system, not the diagram alone.

Routine Maintenance Program — Four Recurring Tasks

Carried out on a fixed schedule, these four tasks prevent almost every major hatch cover failure.

1

Lubrication

Grease all metal moving parts; apply silicon grease-spray to rubber corners and end pieces.

2

Cleaning

Clear hatch tape / foam residue and remove cargo remains from the drain channels.

3

Derust & Paint

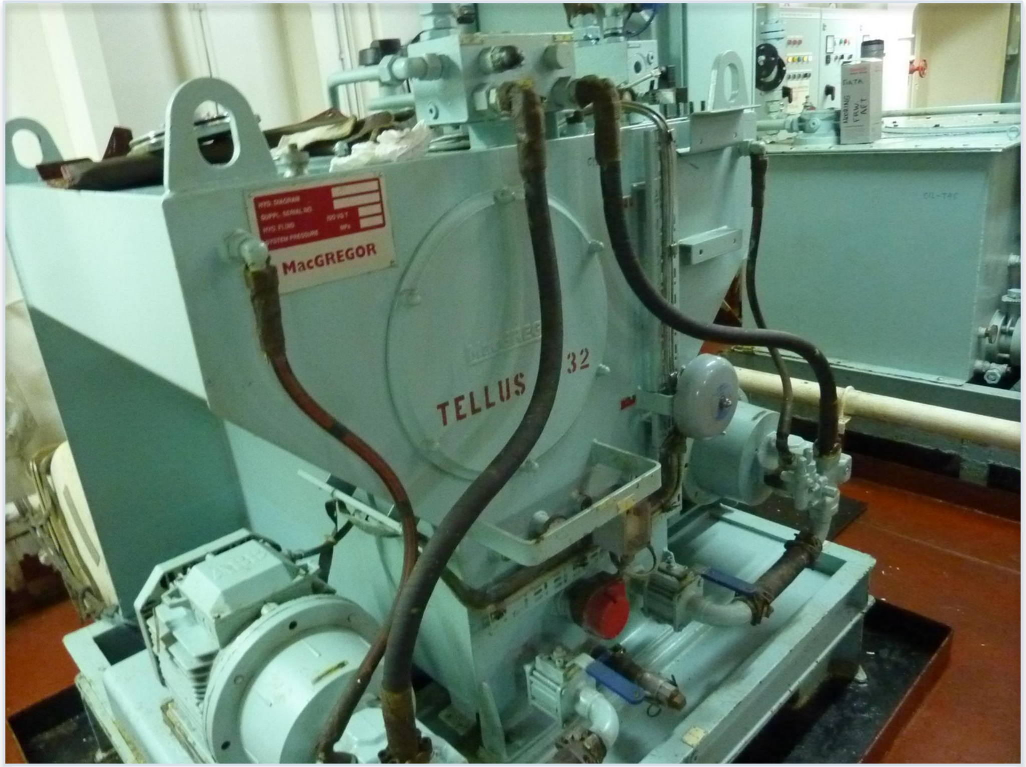
Treat and repaint drain channels and rubber-channel edges before corrosion spreads.

4

Protection & Cleaning

Inspect and protect controls, cylinders and motors from corrosion and contamination.

Step 1 — Lubrication



- Grease all metal moving parts on the fixed schedule, not only when something starts to squeak.
- Apply silicon grease-spray to rubber corners and end pieces to keep the rubber supple and reduce wear from movement.
- A missed lubrication round does not show up immediately — it shows up as accelerated wear weeks later.

Step 2 — Cleaning

Clearing the debris that defeats both sealing and drainage.

- Remove hatch tape and foam residue that accumulates around the seal line.
- Clear cargo remains out of the drain channels — a blocked drain defeats the entire drainage system.
- Cleaning is quick, cheap, and the single most frequently skipped step on this list.

SECTION 08

Step 3 — Derusting & Painting

Protecting the steel that the seal and drainage systems depend on.

- Treat and repaint the drain channels before surface corrosion becomes structural.
- Treat rubber-channel edges the same way — corrosion here undermines the rubber's seating even when the rubber itself is sound.
- Respect drying times before the channel goes back into service, exactly as at drydock.
- A light routine touch-up between drydocks is far cheaper than the full blast-and-paint sequence a neglected channel eventually needs.

Rust never stays confined to where you first see it — treat the drain channel and the rubber channel together, on the same round.

Step 4 — Protection & Cleaning

Closing the loop between mechanical, sealing and hydraulic maintenance.

- Inspect and protect controls, cylinders and motors from corrosion and contamination.
- This step closes the loop between mechanical, sealing and hydraulic maintenance — none of the three groups can be skipped.

Lack of a documented lubrication and cleaning plan is one of the most common causes of premature pad, rubber and hydraulic wear.

Why Prepare Before Drydock

- Time in the yard is expensive — the more that is known before arrival, the less is discovered (and negotiated) after arrival.
- Preparation turns a general 'hatch cover overhaul' into a scoped, priced repair list.
- A well-prepared scope also protects the vessel from unnecessary work the yard may otherwise propose.

Preparation starts long before the vessel enters the yard.

Preparation Checklist

Five checks before the vessel ever reaches the yard.

- **Note** — Record sounds, operating speed and actions during a full cycle before docking.
- **Hydraulic** — Check system pressure and filter condition.
- **UT Test** — Carry out ultrasonic and visual checks of the steel structure.
- **Operate** — Watch the complete operating cycle of every panel.
- **Prepare** — Ask all levels — Master to Bosun — about existing issues.

SECTION 09

Repair Scope — Completion Targets

The completion targets a well-scoped repair should hit.

Scope Item	Target
Pads & stoppers in correct position	90%
Rubber renewed, channel maintained	100%
Mechanical spares replaced	50%
Hydraulic spares replaced	30%

Repair goals are a starting scope for negotiation with the yard — every vessel's condition survey should confirm or adjust these targets.

SECTION 09

Drydock Repair Scope, by Component Group

The same four groups from Section 03, now framed as a repair scope.

Pads & Stoppers — Renewal and welding build-ups where positions are worn low.

Sealing & Drain — Full rubber renewal and drain channel repair — normally the largest single scope item.

Mechanical — Targeted spares for cleats, wheels, hinges and chain, prioritised by condition.

Hydraulics — Controls, seals and hoses replaced selectively rather than as a blanket renewal.

SECTION 09

Interviewing the Crew

The 'Prepare' step means asking, not assuming.

- Ask every level, from Master to Bosun, about issues they have noticed during normal operation.
- Unusual sounds, slow operation, or a panel that needs 'a bit of help' to close are all real defects, even if no one has logged them yet.
- The people who operate the covers every day usually know exactly where the problem is before any inspector arrives.
- Write down what each person says in their own words, and match it against the note taken during the 'Operate' step before the vessel docks.

A crew interview costs nothing and often surfaces the exact defect a full inspection would otherwise take hours to find.

SECTION 09

UT & Visual Testing

Two checks that answer different questions.

- Ultrasonic thickness testing checks the steel structure for wastage that is not visible from the surface.
- Visual checks confirm general condition, but UT is what actually quantifies remaining steel thickness.
- Both are carried out during preparation, not left until the covers are already stripped down.
- Results from both feed directly into the steel-works scope in the drydock repair specification, not into a separate report nobody reads.

A UT reading is only useful if it reaches the person writing the repair scope — route the results there, not just into the file.

SECTION 09

Negotiating Scope With the Yard

Preparation is as much commercial as it is technical.

- Specify the work in detail before asking the yard to price it — a vague scope invites vague, and expensive, quotes.
- List any conflicting works, such as steel replacement or hold blasting, that may interact with the hatch cover repair schedule.
- Negotiate prices against the specified scope, not against a general 'hatch cover overhaul' allowance.
- Agree a change-order process up front for anything found once the covers are stripped down, so mid-repair surprises don't become mid-repair disputes.

The scope you specify before docking is the scope you can actually negotiate — everything found afterwards is priced from a weaker position.

Alignment Procedure

- Remove all perimeter rubber.
- Fit 4 alignment blocks per panel: block height = rubber height minus design compression.
- Close the cover gently and measure resting-pad clearance.
- Do not skip the gentle-close step — forcing the panel shut on new blocks can mask a genuine alignment problem.

SECTION 10

Adjusting Sliding Positioners

Fine-tuning where the panel actually settles.

- Sliding positioners guide the panel into its final resting position as it closes.
- Adjust positioners after pad renewal, not before — the new pad height changes the position the panel actually settles into.
- Confirm smooth engagement through several full close-open cycles before considering the adjustment complete.
- A positioner that is fighting the panel rather than guiding it usually means the pad renewal itself needs re-checking first.

Sliding positioners are a fine-tuning step, not a fix for a pad or alignment problem — chase the root cause before adjusting the positioner.

Ordering Rubber — Key Rules

Five rules that prevent the most common rubber-ordering mistakes.

- Never use generic (IMPA) catalogue lengths — order to the vessel's actual measurements.
- Add 10% to straight rubber lengths, plus 2 extra corner / end pieces.
- Budget 1 kg of glue per 10 metres of rubber.
- Use Sikaflex 221 / 198i and Loctite 424 / 480 — not a generic substitute.
- Do not rely on onboard stock; check rubber quality on receipt.

SECTION 10

Rubber Channel Preparation

Getting the channel ready before any new rubber goes in.

- Remove all old rubber and glue completely — blasting alone will not remove glue and rubber residue.
- Sandblast the rubber channel and the resting pads together.
- Repair any rusted steel areas found during blasting before proceeding.
- Apply a full paint coat and respect the manufacturer's drying times before installing new rubber.
- Inspect the finished channel profile before ordering the final rubber lengths — steel repairs can shift the exact dimensions slightly.

Every shortcut taken in channel preparation shows up later as a rubber piece that will not seat or bond correctly.

Installing Rubber — Key Rules

Six rules for a rubber installation that lasts to the next drydock.

- Fit corner and end pieces first, trimmed to align exactly with the compression bar.
- No straight piece shorter than 2 metres.
- Apply glue generously, especially at the sides of the channel.
- Install with the natural hills and valleys of the rubber profile.
- Wait 24 hours before closing the cover.
- Finish corners with Sikaflex.

Rubber & Glue — Getting the Bond Right

The bond matters as much as the rubber piece itself.

- Glue and rubber together form the bond — a good rubber piece with a poor glue job still fails.
- Apply glue generously, especially at the sides of the channel, and respect the maker's open time before fitting.
- A rubber piece that lifts at the edge within the first weeks almost always points back to the glue application, not the rubber itself.

Workshop 1D — Calculating Total Rubber for a Drydock

A full worked example with the real numbers from a second-vessel drydock.

WORKSHOP EXERCISE

- Calculate the total rubber required for a drydock on a second vessel, covering both cross-joint and perimeter lengths across all panels.

Worked Solution

3,220 × 4 + 5,555 × 4 + 17,934 × 5 = 124.77 m

(3,835 × 4 + 6,029 × 4 + 17,934 × 5) × 4 = 516.50 m

Total rubber required: 641.27 m

Add 10% ordering margin: 705.10 m

Rubber ordered: 710 m

Wheel Repairs

- Check freeplay first — excess play is usually the earliest sign of wear.
- Check the flange for wear or damage that could let the wheel ride off the track.
- Decide whether the wheel track itself also needs repair — a new wheel on a worn track will wear again quickly.

SECTION 11

Hinge Repairs

- Grease all points — this cannot be overstated, it is the single most effective preventive action.
- Check for clearance during operation and listen for noise that signals early binding.
- Intermediate hinges are normally serviced with covers on shore; end hinges on site with the cover closed; cylinder hinges need the same attention as structural hinges.
- Discussion: consider why a hatch cover might stop opening fully — hinge condition is a common answer.

Removing Hinge Pins

- Pins seize because the spherical bearing inside wears and corrodes, gripping the pin far tighter than its running clearance would suggest.
- Options for a stuck pin: weld a nut onto the pin, screw in a large bolt, or use a special hydraulic extraction cylinder together with heating.
- Even with the right method, expect roughly a 70% success rate — some pins will still require the alternative approach.

Cleat Repairs

A short, strict sequence — with one absolute rule.

- Clean thoroughly — degrease and derust before assessing true condition.
- Check the nut condition specifically, not just the cleat body.
- Replace worn parts — never weld a cleat component back into service.
- Replace steel washers and rubber washers together as a set.

SECTION 11

Chain Repairs

- Replace chain once wear exceeds 20% of thickness, or once correct tensioning can no longer be maintained.
- Check all chain wheels for wear at the same time as the chain.
- Confirm there is no twist and no slack in the chain run.
- Tension the chain with the cover in the closed position before jacking down.

SECTION 11

Lifting Blocks

Small components that get very little attention until they bind.

- Check alignment of lifting blocks before every major lift operation.
- Confirm sliding clearance is within tolerance — a tight block binds under load, an over-loose block allows uneven lifting.
- Inspect the block and its guide together, not the block alone — most binding problems start in the guide, not the block.
- Include lifting blocks in the same drydock scope as wheels and hinges — they see comparable loads and are just as easy to overlook.

A binding lifting block on a heavy panel is a crane-load safety issue, not a minor mechanical inconvenience.

SECTION 11

Panel Steel Works

Cutting, fitting and welding — the sequence matters.

- Cutting: remove only the damaged or worn steel identified by the UT survey, not an arbitrary panel area.
- Fitting: dress and fit the replacement piece accurately before any welding begins.
- Welding: follow the correct sequence and pre-heat requirements for the steel grade to avoid distortion.
- Recheck panel flatness and pad-line alignment after welding, before the panel goes back to the rubber and pad renewal stages.

Steelwork done out of sequence is the single most common reason a freshly repaired panel still doesn't sit flat afterward.

SECTION 11

Fair-Up Procedure

Straightening a distorted panel without replacing it.

- The hot side is the short side — heat the side of the panel you want to expand relative to the other.
- Use hydraulic pressure together with heating rather than heat alone for controlled correction.
- Heat along the stiffeners and cool down with water to fix the correction in place.
- Work in small, checked increments — fairing a panel is a repeatable procedure, not a one-shot correction.

Fair-up is a skilled steelwork task in its own right — it belongs with the same supervision as any other structural repair on the panel.

Rubber Channel Repair — Sequence

A specific order that avoids re-work.

- Replace the outer side of the channel first.
- Use a wood guide for fitting the inner plate accurately.
- Confirm the finished channel profile against the rubber it will receive before final welding.
- Coordinate this repair with the rubber channel preparation stage in Section 10 — the two are one job, done by two different trades.

A channel that is welded slightly out of profile will fight the rubber for the life of that rubber — check profile before, not after, final welding.

SECTION 11

Hydraulic Pipe Repairs

Five points that separate a clean repair from a contaminated system.

- Drain oil using air, not by simply opening the line, to avoid an uncontrolled spill.
- Fit plugs to isolate the section being worked on.
- Replace long sections rather than patching repeatedly along the same run.
- Renew flanges and fit the newer type of pipe supports where available.
- Cleanliness during the repair is not optional — contamination introduced here travels through the entire system.

Controls Overhaul

- Decide whether each valve can be reconditioned or needs replacement.
- Renew O-rings as standard practice on any valve that is opened up.
- Trace leaks from control levers back to their source before closing up the job — a re-greased lever that still leaks has not been repaired.

Cylinder Gasket Set Explained

- A full gasket set covers the O-ring, oil seal, backup ring and scraper (dust) seal.
- Regasket on every overhaul as standard practice, regardless of how the old seals look.
- Confirm whether heating is permitted for opening the cylinder before applying it — not every cylinder tolerates the same approach.

Cylinder Examination

- Check for residue inside the cylinder that indicates contamination or seal breakdown.
- Inspect the piston surface for scoring or wear that would damage a new seal on reassembly.
- Carry out a seal check before deciding the full extent of repair needed.

Cylinder Repair

- Rechrome the rod only when surface condition requires it, not as a routine step.
- Regasket always, and reposition the dust seal correctly on reassembly.
- Grinding is sometimes necessary to restore a true surface — confirm before skipping it.
- Test both sides of the cylinder after reassembly, not just the side that was repaired.

Pump Unit Checks

- Check pressure and amperage together — a mismatch between the two is an early warning sign.
- Check oil level and oil temperature during operation.
- Confirm when the tank was last cleaned and the filter last changed, and log it.

SECTION 12

Hydraulic Motors — Checks

Two checks, done consistently, prevent most motor failures.

- Check for leaks at the casing and shaft seal.
- Replace the shaft seal as soon as weeping is visible, rather than waiting for a full leak.
- Listen and feel for unusual heat or vibration during a normal operating cycle, in the same way as for a hydraulic motor anywhere else on board.
- Log every check against the specific motor position — a fleet of identical-looking motors can still wear at very different rates.

A hydraulic motor gives very little warning between 'slightly weeping' and 'losing drive' — treat the first sign as the deadline to act.

Drydock Usual Schedule — Six Stages

The sequence a typical hatch cover drydock repair follows, start to finish.

1
Preparation
Inspect, order spares, specify the work scope and negotiate yard prices.

2
Remove Rubber
Strip all rubber and worn resting pads from every panel.

3
Align Hatches
Use plastic blocks to align panels; weld new pads as needed.

4
Install Rubber & Components
Fit new rubber plus new mechanical and hydraulic components.

5
Adjustment
Adjust chain and sliding positioners to bring panels into correct position.

6
Final Testing & UT
Complete final testing and ultrasonic thickness testing before sail-away.

Full Drydock Repair Cost Breakdown

Every line item from a real 10-year drydock repair budget — nothing grouped or omitted.

Repair Item	Cost (USD)
New Rubber	$10,000
New Pads	$4,000
Rubber Installation	$10,000
Blasting & Painting Rubber Channel	$6,000
Steel Works (Rubber Channel Damages)	$5,000
Pads Welding & Build-Ups	$5,000
Hose & UT Test	$2,000
Mechanical Spares (Cleats, Wheels)	$2,000
Hydraulic Spares (Controls, Seals, Hoses)	$5,000
Total (10-Year Repair Budget)	$49,000

SECTION 13

Hidden Costs

Two items the schedule and headline budget do not always show.

- Pipe replacement is frequently under-scoped — once pipework is opened up, the true extent of chafe damage is often larger than the visual inspection suggested.
- Oil is a real, recurring cost across the whole hydraulic system — full system changes and top-ups during a drydock scope add up beyond the headline spares figure.
- Neither item has its own line in the standard nine-item budget on the previous slide — both are usually absorbed into 'steel works' or 'hydraulic spares' until they grow large enough to notice.
- Ask the yard to itemise pipe and oil costs separately in the quotation, rather than folding them into a general allowance.

Hidden costs are hidden only because nobody asked for them by name — put them on the scope sheet and they stop being a surprise.

SECTION 13

Expert Supervision — Five Benefits

What independent technical supervision actually delivers during a drydock repair.

Cooperation

Planning of works for proper results between owner and yard.

Responsibility

Correct decisions taken on the owner's side throughout the repair.

Productivity

Coordinated crew operations and shipyard repairs, run efficiently.

Money Savings

Deciding on work priority and cancelling work that does not fit the schedule.

Client Satisfaction

Final responsibility for the outcome resting with one accountable party.

Workshop 1E — Two Field Problems

Two real diagnostic questions for open discussion.

WORKSHOP EXERCISE

- When closing folding hatch covers, the oil tank overflows. What is the problem?
- When opening folding hatch covers, the safety hook cannot be engaged. What is the problem?

SECTION 13

Cost Control Tips

Turning the last five slides into fewer surprises at the next drydock.

- A specified, itemised scope prices far more predictably than a general allowance.
- Independent supervision pays for itself by cancelling work that does not fit the schedule, not only by finding defects.
- Track hidden-cost items like pipe replacement and oil separately so they don't quietly erode the headline budget.
- Compare actual spend against the nine-item budget after every drydock, not just before it, so the next scope is built on real figures.
- A repair history that tracks cost per item, not just total cost, is what lets the next preparation stage negotiate from real data instead of guesswork.

The cheapest drydock is rarely the one with the lowest quoted price — it's the one with the fewest items discovered after the ship is already docked.

SECTION 14

Common Mistakes — Overview

One finding matters more than any item on the three lists that follow.

- Ask for assistance from experts.
- Lack of supervision during repairs is the single biggest cause of recurring hatch cover problems — more so than any individual technical mistake on the following pages.

SECTION 14

Weathertightness Mistakes

Five ways a rubber renewal quietly fails.

- Replace rubber without fixing the pads.
- Install backing-strip rubber everywhere as a shortcut.
- Mix new and old rubber together.
- Fill gaps with silicone instead of the correct rubber piece.
- Skip or under-paint the rubber channel.

Every one of these five looks like a shortcut at the time and shows up as a re-work item at the next drydock.

SECTION 14

Mechanical Mistakes

Three habits that let small defects become big ones.

- Ignore abnormal sounds during operation.
- No greasing plan or fixed schedule.
- Attempt on-board repairs instead of proper shore repair.

Every one of these three is free to fix and expensive to ignore — none of them require a drydock to correct.

SECTION 14

Hydraulic Mistakes

Five habits that damage the system faster than normal wear.

- Clean the filter instead of changing it.
- Close covers without running the pump.
- Change pipes without flushing the system.
- Leave valves in the wrong position during the voyage.
- Ignore leaks.

None of these five save real time — they only move the cost from a scheduled maintenance task to an unscheduled breakdown.

Questions for Group Discussion

Five real questions the masterclass leaves open — discuss them as a crew.

- Should the bearing pads be greased?
- Should the rubber packing be greased?
- Should the pipes be covered with petrotape?
- Should hatch tape be used?
- Should foam be used?

There is more than one defensible answer to each of these — confirm what your vessel's maker and CSM actually specify before deciding as a crew.

Key Takeaways

Six things every crew member must remember from this masterclass.

Weathertightness ≠ Watertightness

Sealing depends on pads, rubber compression and drains — never assume weight or cleats alone are enough.

Follow the Maintenance Plan

Lubricate, clean, derust and protect on a fixed schedule — most failures start as a skipped routine task.

Renew Rubber Correctly

Order to spec, align with the compression bar, and wait 24 hours before closing the cover.

Inspect Mechanical Components

Check wheel freeplay, hinge clearance, cleat condition and chain wear on every round.

Respect Hydraulic Discipline

Never close covers without the pump running, and never ignore a leak.

Ask for Expert Supervision

Independent technical oversight during drydock repairs is the biggest safeguard against recurring problems.

SECTION 14

Crew Assessment Questions

Use these to confirm understanding before signing off the training record.

1. What determines whether a hatch cover is weathertight — mechanical clamping, or pad and rubber compression?
2. What is the difference between a side-rolling, folding and lift-away hatch cover?
3. How is the total force on a hatch cover's perimeter rubber calculated?
4. What is the correct sequence for renewing rubber, from channel preparation to closing the cover?
5. When must a chain be replaced during inspection?
6. Why can a hinge pin be difficult to remove, and what methods are used to extract one?
7. What are the four items in a complete cylinder gasket set?
8. What targets should a well-scoped drydock repair aim to complete for pads, rubber, mechanical and hydraulic spares?
9. What is the single biggest cause of recurring hatch cover problems?

Training Complete

Hatch Cover Operations & Maintenance

Training records must be completed and signed by the Chief Officer following each session.